MIL-D-OO10132K(EA)
17 December 1981
SUPERSEDING
MIL-D-OO10132J(EA)
22 February 1977
USED IN LIEU OF
MIL-D-10132C
30 September 1964

MILITARY SPECIFICATION

DISKS, VALVE

This limited coordination military specification has been prepared by the US Army Armament Research and Development Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-D-10132C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers natural ribber disks for use as components of protective mask valves.

1.2 Classification. The valve disks shall be of the following types:

Type II - Disk, Valve, Nonmetallic, Outlet (Drawing B5-2-1006).
Type III - Disk, Valve, Nonmetallic, Resuscitator (Drawing D5-81-150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified (see 6.2), the following Standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use In improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-289 - Visual Inspection Guide for Rubber Sheet Material.
MIL-STD-298 - Visual Inspection Guide for Rubber Extruded Goods.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D5-2-606		Seat, Outlet Valve.
B5-2-1006		Disk, Valve, Nonmetallic, Outlet.
D5-81-3		Voicemitter-(Outlet Valve, C29R1.
E5-81-125		Voicemitter-Outlet Valve, C29R5.
D5-81-134		Valve Seat Housing Assembly.
D5-81-150		Disk, Valve, Nonmetallic, Resuscitator.
B136-44-1-1	-	Resistance Indicator, Inlet and Outlet Valve, Q212, Assembly (for C29R1 Voicemitter-Outlet Valve).
B136-44-1-3	-	Resistance Indicator, Inlet and Outlet Valve, Q212, Assembly (for C15 Outlet Valve).
B136-44-1-4	-	Resistance Indicator, Inlet and Outlet Valve, Q212, Assembly (for C29R5 Voicemitter-Outlet Valve).
D136-45-202-3	-	Indicator, Outlet Valve Leakage, M4A1, Assembly (for C29R1 Voicemitter-Outlet Valve).
D136-45-202-14	-	Indicator, Outlet Valve Leakage, M4Al, Assembly (for C15 Outlet Valve).
D136-45-202-15	-	Indicator, Outlet Valve Leakage, M4A1, Assembly (for C29R5 Voicemitter-Outlet Valve).

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of solicitation shall apply.

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ASTM STANDARDS

D412 - Rubber Properties in Tension.
D572 - Rubber - Deterioration by Heat and Oxygen.
D573 - Rubber - Deterioration in an Air Oven.
D2240 - Rubber Property - Durometer Hardness.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The robber compound shall be nonblooming and shall contain no less than 92 percent by volume new natural rubber. A slight paraffin bloom shall not be considered a violation of the nonblooming requirement of this specification.

3.1.2 Antioxidant. The rubber compound shall contain an antioxidant.

3.1.3 Copper and manganese inhibitor. The rubber compound shall contain a copper and a manganese inhibitor.

3.2 Manufacture. The disks shall be manufactured by the conventional process of molding under heat and pressure in accordance with Drawing B5-2-1006 for type 11 or Drawing D5-81-150 for type III.

3.2.1 Test slabs. Fourteen test slabs, 6 by 6 by 0.035 ± 0.012 inches, shall be made from the preproduction sample stock and from the stock used for each regular production lot. The slabs shall have a cure equivalent to that of regular production disks (see 6.3).

3.3 Physical properties.

3.3.1 Unaged.

3.3.1.1 Tensile strength. The average tensile strength of the test slab specimens shall be no less than 17.9 megapascals when tested as specified in 4.4.4.1(c).

3.3.1.2 Ultimate elongation. The average ultimate elongation of the test slab specimens shall be no less than 650 percent when tested as specified in 4.4.4.1(c).

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3.3.1.3 Tensile set. The average tensile set shall not exceed 15 percent when specimens are stretched 600 percent, held for 10 minutes, released, and rested-for 10 minutes when tested as specified in 4.4.4.1(c).

3.3.1.4 Durometer hardness. The average Type A Shore Durometer hardness shall be 40 ± 5 for the type III disk when tested as specified in 4.4.4. l(d).

3.3.2 Aged (oven process).

3.3.2.1 Tensile strength and ultimate elongation. After aging in accordance with 4.4.4.l(b), the average tensile strength and average ultimate elongation of the test slab shall increase or decrease no more than 20 percent of actual values obtained in 3.3.1.1 and 3.3.1.2 when tested as specified in 4.4.4.1(c) The tensile strength shall not be less than 17.9 megapascals.

3.3.2.2 Disk characteristics. After aging in accordance with 4.4.4.l(b), the disks shall exhibit no tackiness or increased hardness. The disks shall not tear when tested as specified in 4.4.4.2(a).

3.3.3 Resistance to oxygen-pressure aging. Disks shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure of 2.10 ± 0.10 megapascals and a temperature of 70° + l°C, when tested as specified in 4.4.4.3.

3.4 Resistance to airflow. When the outlet valve Is clean, thoroughly dry, and properly assembled to the outlet valve seat as shown on Drawing D5-2-606 (Seat. C15 Outlet Valve), D5-81-3 (C29R1 Voicemitter-Outlet Valve), or E5-81-125 (C29R5 Voicemitter-Outlet Valve Assembly), whichever is applicable, the resistance to airflow through the assembled outlet valve less valve seat housing assembly (Drawing D5-81-134) shall not exceed 22 millimeters of water at an airflow of 85 liters per minute when tested as specified in 4.4.4.2(b). When testing the complete 0C29R5 voicemitter assembly, resistance to airflow shall not exceed 30 millimeters of water at an airflow of 85 liters per minute when tested as specified in 4.4.4.2(b).

3.5 Leakage. When the outlet valve disk is thoroughly dry and properly assembled to the outlet valve seat as show on Drawing D5-2-606 (Seat, C15 Outlet Valve), D5-81-3 (C29R1 Voicemitter-Outlet Valve), or ES-81-125 (C29R5 Voicemitter-Outlet Valve Assembly), whichever is applicable, the leakage through the assembled outlet valve shall not exceed 15 milliliters of air per minute at a pressure drop of 1.0 inch of water when tested as specified in 4.4.4.2(c).

3.6 Preproduction. Prior to the start of regular production and followlng any change in regular production methods or rubber formulation, a preproduction sample of disks and test slabs shall be produced in accordance with this specification for examination and test (see 4.3).

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3.7 workmanship. The disk surfaces shall be smooth and free from imperfections, such as porous areas, cracks, checks, flash, abrasions, and blisters. The rubber shall be compact and free from holes, bubbles, foreign matter, and tackiness. Ridges of mold parting lines, when closely trimmed, shall not be considered as flash. Rips, tears, or punctures shall be considered as damage. For minor visual defects, the visual inspection standards of MIL-STD-289, MIL-STD-298 and MIL-STD-407 shal1 apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Mold inspection. Prior to initiation of the preproduction sample, the contractor shall submit to the Government the actual inspection data that shows that each applicable mold meets the requirements of the applicable drawing.

4.3.2 Sample. A preproduction sample of 200 disks and 14 test slabs (see 3.2) shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Preproduction sample disks shall be inspected for compliance with the drawings and the requirements of this specifications

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4.3.3.2 For teats. Fourteen test slabs shall be tested in accordance with 4,4.4.1. TWO groups of 35 disks each shall be tested in accordance with 4.4.4.2(a), disk characteristics and 4.4.4.3, resistance to oxygen pressure aging.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.1, 4.3.3.1, and 4.3.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A regular production lot shall consist of the valve disks of the same type produced by one manufacturer from one batch of compounded raw material under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Physical properties Fourteen test slabs, prepared in accordance with 3.2.1, shall be submitted with each lot for testing.

(b) Disk characteristics. Thirty-five representative disks shall be taken from each production lot.

(c) Resistance to oxygen-pressure aging. Valve disks shall be taken in accordance with MIL-STD-105, inspection level S-4; however, in no case shall the sample size be less than 50 disks.

4.4.3 Inspection procedure-

4.4.3.1 For examination and nondestructive tests. Sample disks from regular production lots shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests.

(a) Physical properties. Sample test slabs shall be tested in accordance with 4.4.4.1. The acceptance number is zero.

(b) Disk characteristics. Sample disks shall be tested as specified in 4.4.4.2(a). The acceptance number is zero.

(c) Resistance to oxygen-pressure aging. Sample disks shall be tested in accordance with 4.4.4.3. The acceptance number is zero.

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4.4.3.3 Classification of defects.

(a) Disk, valve, nonmetallic, outlet (Drawing B5-2-1006).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.25 percent defective	
101	Resistance to airflow	4.4.4.2(b)
102	Leakage	4.4.4.2(c)
	AQL 1.0 percent defective	
103	Thickness (0.028) incorrect	
104	Workmanship (3.7 except Minor 203)	
105	Depth (0.182) incorrect	
106	Thickness (0.024) incorrect	
Minor:	AQL 2.5 percent defective	
201	Marking missing or incorrect	
202	Stem length (rein) undersize	
	AQL 4.0 percent defective	
203	Minor defects listed in MIL-STD-289, MIL-STD-298 and MIL-STD-407 (see 3.7)	

(b) Disk, valve, nonmetallic, resuscitator (Drawing D5-81-150).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Thickness (0.016) incorrect
102	Width dimension (0.161) incorrect
103	Workmanship (3.7 except Minor 202)
Minor:	AQL 4.0 percent defective
201	Stem length incorrect
202	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407 (see 3.7)

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(c) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	None defined
Minor:	AQL 4.0 percent defective
201	Creases in fiberboard material after insertion in box

4.4.4 Tests.

4.4.4.1 Test slabs.

(a) Test specimens. Cut thirty-five dumbbell specimens from the test slabs in accordance with ASTM D412. Do not buff or stretch the specimens before testing. Test ten specimens before aging and ten specimens after oven aging for tensile strength and ultimate elongation. Test five specimens before aging for permanent set.

(b) Aging (oven process). Age the test specimens selected for oven process aging in an air oven at a temperature of 100° ± l°C for 24 hours continuously as specified in ASTM D573.

(c) Tensile strength, ultimate elongation, and tensile set. Determine tensile strength, ultimate elongation, and tensile set tests of the unaged and aged specimens as specified in ASTM D412. Record actual test values.

(d) Durometer hardness. Determine the durometer hardness as specified in ASTM D2240.

4.4.4.2 Disk.

(a) Disk characteristics. Age each of the 35 sample disks as specified in 4.4.4.l(b) and examine for tackiness and hardness. For type II disks, make two gage marks 1/2 inch apart on the periphery of the disk. Grasp the disks with the fingers at the gage marks, stretch until the gage marks are 1-1/2 inches apart and permit the disk to return to its original unstressed condition. Examine the disk for tears. For type III disks, make two gage marks 1/2 inch apart on the diameter of the disk (each mark shall be 1/4 inch from the center of the disk). Grasp the disk with the fingers at the gage marks, stretch until the gage marks are 1 inch apart and permit the disk to return to its original unstressed condition. Examine the disk for tears.

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(b) Resistance to airflow. Determine the resistance to airflow with Q212 Inlet and Outlet Valve Resistance Indicator Assembly (Drawing B136-44-1-3, B136-44-1-1, or B136-44-1-4, as applicable).

(c) Leakage- Determine leakage with M4A1 Outlet Valve Leakage Indicator Assembly (D136-45-202-14, D136-45-202-3, or D136-45-202-15, as applicable) l

4.4.4.3 Resistance to oxygen-pressure aging Test the disks in accordance with ASTM D572.

59 PACKAGING

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The disks shall be packaged and packed to provide adequate protection from physical damage from the supply source to.the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractor's name, address, and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. The type 11 disk is intended for use as a component of the C15 Outlet Valve, C29R1 Voicemitter-Outlet Valve Assembly, and C29R5 Voicemitter-Outlet Valve Assembly. The type 111 disk is intended for use as component of C29R5 Voicemitter-Outlet Valve Assembly.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of disk and color.
(c) The valve seats for conducting tests will be furnished by the contracting officer.
(d) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

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6.3 Equivalent cure. Slabs and disks of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free sulfur contents do not differ by more than 10 percent
(b) Their curing times can be equated as follows:

$$t_2 = t_1 \times 1.6^{(T_1 - T_2)/10}$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature °F of cure for time t_1, and
T_2 = temperature °F of cure for time t_2.

For convenience, the nomogram in figure 1 may be used instead of the mathematical computation in determining the equivalent cure.

6.4 Interplant shipment. Packaging for supplies and materials that will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 214, 216, 218, and 234. When amendment, revision or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices, to change the agreement or make other appropriate accommodations.

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Preparing activity:

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Project No. 4240-A768

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EQUIVALENT CURE TIME FOR NATURAL RUBBER

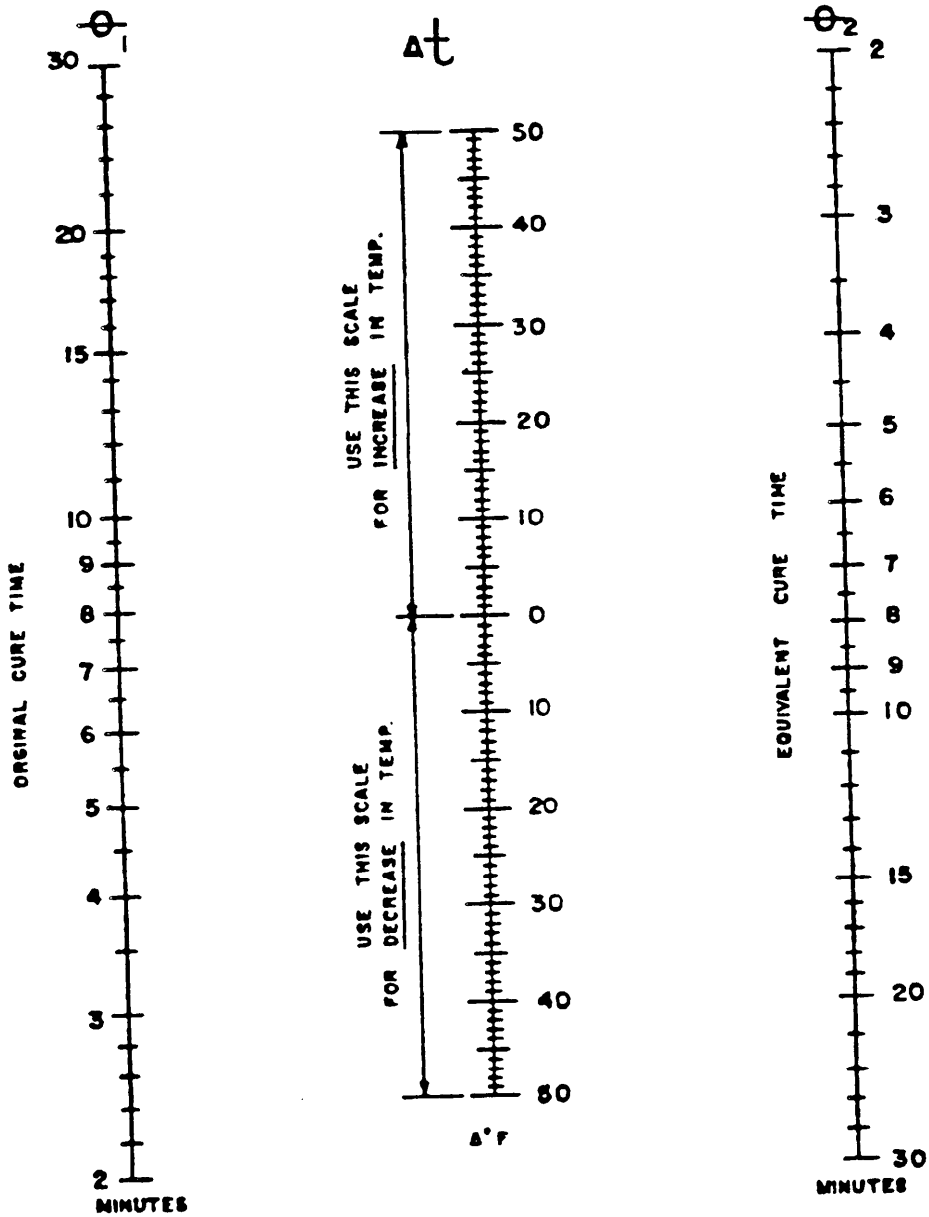


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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